From exposure genetics to clinical translation in lung cancer

Hypothesis-driven framework integrating causal inference, multi-omics validation, and prognostic modeling

Exposure GWAS
30 metabolic, inflammatory,
and lifestyle traits

Outcome GWAS
overall lung cancer,
LUSC, and LUAD

01

Causal screening

UVMR + MVMR
Bidirectional MR design
Sensitivity analysis and pleiotropy control

02

Bayesian colocalization

PP.H4 evidence stratification
SNP-gene-disease linking
Prioritized candidate drivers

03

Multi-omics validation

TCGA transcriptome + methylome
CPTAC proteome evidence
Cross-layer biological concordance

04

Clinical translation

LASSO-Cox risk score construction
External validation (GSE39279, GSE30219)
KM + Cox + ROC performance

Output

Causal pathways linked to lung cancer susceptibility
Multi-omics biomarkers translated into prognostic stratification